

The logo features the word "RAOH" in a large, bold, white sans-serif font. The background is a dark teal color with abstract, lighter teal geometric shapes, including a large circle on the left and a series of curved lines at the bottom right.

RAOH

— R e s e a r c h e r s —

RESEARCH. DISCOVER. IMPACT.

Advancing knowledge.
Creating real-world impact.

Medical Research Study Designs

Understanding How Medical Evidence Is Built

Every medical breakthrough begins with a question.
Why do some people develop a disease while others do not?
Does a new treatment work better than the current one?
Can a lifestyle habit increase the risk of illness?

Medical research is the process of finding reliable answers to these questions through systematic investigation rather than assumptions, opinions, or personal experiences.

Think of medicine as a vast puzzle.

Each research study represents a different method of collecting pieces of evidence. Some studies simply observe what happens in the real world, while others actively test interventions and compare outcomes. Some look back into the past, whereas others follow participants into the future.

Because no single study can answer every question, researchers have developed multiple study designs, each with its own purpose, strengths, limitations, and level of scientific evidence.

Understanding these designs is one of the most important foundations of evidence-based medicine.

What Is Research?

Research is a structured scientific process used to generate new knowledge, verify existing information, and answer specific questions using evidence.

Instead of relying on intuition or assumptions, researchers:

- Ask a clear question
- Collect data systematically
- Analyze the findings
- Draw evidence-based conclusions

Example

Suppose we want to answer the following question:
Does sleep deprivation affect medical students' academic performance?

There are several ways to investigate this question, and the choice of study design determines how the evidence will be collected and interpreted.

This is where research methodology begins.

The Two Major Categories of Medical Studies

Medical research is generally divided into two broad categories:

1. Observational Studies

In observational studies, researchers do not interfere with participants.

They simply observe individuals, measure variables, and record outcomes as they naturally occur.

Think of standing on the sidelines of a football match and taking notes without influencing the game.

These studies are widely used because they are often easier, faster, and more practical than experimental studies.

2. Experimental Studies

In experimental studies, researchers actively intervene.

They may administer a treatment, introduce an exposure, assign participants to groups, or test a new intervention.

Instead of merely observing events, researchers become part of the process.

These studies provide some of the strongest evidence available in medicine because they allow direct evaluation of cause-and-effect relationships.

Observational Study Designs

Observational studies form the backbone of clinical and epidemiological research.

All share one key characteristic: the researcher observes but does not intervene.

Case Report

A Case Report describes a single patient with an unusual disease, unexpected outcome, novel treatment response, or rare clinical finding.

Importance

Many major medical discoveries first appeared as individual case reports.

Example

A physician encounters a patient who develops an extremely rare adverse reaction to a newly introduced medication. The physician documents:

- Clinical presentation
- Diagnostic findings
- Treatment course

- Outcome
- Lessons learned

Strengths

- Identifies new phenomena
- Generates hypotheses
- Useful for rare conditions

Limitations

- Single patient only
 - Cannot establish causation
 - Not generalizable
-

Case Series

A Case Series expands a case report by describing multiple patients with similar characteristics.

Example

Ten patients experience the same unexpected complication after a medication. Patterns are analyzed across all cases.

Strengths

- Identifies trends
- Stronger than single case report

Limitations

- No control group
 - Cannot determine risk or causality
-

Cross-Sectional Study

A Cross-Sectional Study provides a snapshot of a population at a specific point in time.

Think of it as taking a photograph.

Example

A survey of 1,000 medical students to determine prevalence of poor sleep quality.

Main Outcome

Prevalence (existing cases at a single time point)

Strengths

- Fast
- Cost-effective
- Good for prevalence
- Easy to conduct

Limitations

- Exposure and outcome measured simultaneously
 - Cannot determine temporality
 - Cannot prove causation
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Case-Control Study

A Case-Control Study starts with disease status and looks backward to identify exposures.

Structure

- Cases: individuals with disease
- Controls: individuals without disease

Example

Investigating smoking and lung cancer by comparing smoking history in both groups.

Key Measure

Odds Ratio (OR)

Strengths

- Good for rare diseases
- Efficient and fast
- Requires fewer resources

Limitations

- Recall bias
 - Selection bias
 - Cannot calculate incidence
-

Cohort Study

A Cohort Study begins with exposure and follows participants over time to observe outcomes.

Example

Following smokers and non-smokers over 10 years to assess lung cancer development.

Types

- Prospective cohort: follows forward in time
- Retrospective cohort: uses past records

Key Measure

Relative Risk (RR)

Strengths

- Establishes temporality
- Measures incidence
- Stronger evidence for causation

Limitations

- Time-consuming
- Expensive
- Loss to follow-up

The Golden Difference

Case-Control:

Disease → Past Exposure

Cohort:

Exposure → Future Disease

Experimental Studies

Experimental studies actively test interventions and provide high-level evidence.

Randomized Controlled Trial (RCT)

The gold standard for evaluating medical interventions.

How it works

- Random assignment into groups
- Group A receives intervention
- Group B receives placebo or standard care
- Outcomes are compared

Key Concepts

Randomization: reduces selection bias

Blinding: reduces performance and detection bias

- Single-blind: participants unaware
- Double-blind: both participants and researchers unaware

Strengths

- Strongest for causality
- Minimizes bias
- High internal validity

Limitations

- Expensive
- Ethical constraints
- Complex logistics

Evidence Synthesis

Systematic Review

A structured method to identify, evaluate, and summarize all relevant studies on a topic using predefined criteria. **(There is a detailed explanation in SRMA section)**

Meta-Analysis

A statistical technique that combines results from multiple studies into a single quantitative estimate.

Produces:

- More precise estimates
- Higher statistical power
- Stronger evidence

Narrative Review

A Narrative Review is a non-systematic overview of literature on a topic.

Characteristics

- Flexible structure
- No strict protocol
- Author-driven selection of studies

Strengths

- Broad overview
- Useful for education and background understanding

Limitations

- Higher risk of bias
 - Not reproducible like systematic reviews
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Editorial and Letter to the Editor

1. Editorial

An **Editorial** is a short scientific article written usually by journal editors or invited experts.

Purpose

- Provide commentary on topics in the journal issue
- Highlight importance of research findings
- Offer expert interpretation or perspective

Key Features

- Not original research
 - Opinion-based but evidence-informed
 - Published alongside journal issues
 - Written by editors or experts
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2. Letter to the Editor

A **Letter to the Editor** is a short academic communication sent by researchers or readers to the journal.

Purpose

- Comment on a published study
- Critique methodology or findings
- Add additional insights or data
- Correct errors in published work

Key Features

- Very short (usually 300–800 words)

- Limited references
 - Focused on one issue
 - May trigger scientific discussion
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Evidence Pyramid

1. Case Report
 2. Case Series
 3. Cross-Sectional Study
 4. Case-Control Study
 5. Cohort Study
 6. Randomized Controlled Trial (RCT)
 7. Systematic Review
 8. Meta-Analysis
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Key Research Terminology

Bias, confounding, population, sample, statistical significance.

Choosing Study Design

- Prevalence → Cross-sectional
 - Rare disease → Case-control
 - Risk over time → Cohort
 - Treatment → RCT
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Final Integrated Example

Research question:

Do energy drinks affect sleep quality?

- Cross-sectional → measure both at once
- Case-control → compare sleep disorder vs no disorder
- Cohort → follow users vs non-users
- RCT → assign exposure and compare outcomes

Each design contributes different evidence to the same scientific question.
